

10 Modbus Read, Write and Override Parameters plus Exception Codes

10.1 Overview

Introduction

This section contains information concerning Reading, Writing, and Overriding parameters in the UDC3200 Process Controller. There are two types of parameters:

- **Data Transfer**—These parameters include reading control data, option status, and reading or changing setpoints.
- **Configuration Data**—All the configuration data is listed in the order in which it appears in the controller.

Each type of parameter has the identifying codes listed with it.

What's in this section?

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General Information

Non-volatile Memory Retention

- This controller uses non-volatile memory to store configuration data. These memories are guaranteed to retain data for a minimum of ten years as long as the data is not written and erased more than 10,000 times. In order not to exceed this number, it is strongly recommended that configurations which change rapidly such as Computer Setpoint use the Override feature which does not affect non-volatile memory.

Analog Parameters

- Whenever analog register addresses 0001 through 0074 (those that can be changed via communications) are changed, a Write cycle occurs after receipt of the message and the response is returned.

Override Parameters

- Override analog register address 007D (computer setpoint) is not stored in non-volatile memory. It can be changed as frequently as desired with no effect on non-volatile memory retentivity, but the controller must remain in the slave mode.

Digital Parameters

- Whenever digital configuration register addresses 0080 through 00FA are updated via communications, the non-volatile memory is updated as soon as the message is received.

Communications Transfer Rates

- Reads minimum 20mS and writes minimum 200mS

Supported Function Codes

- IR port 20 and 21
- RS485 and Ethernet ports 1,2,3,4,6,16,17,20,21

Communications Modes of Operation

- When the Shed Timer is enabled and a write or override occurs the controller will enter Slave Mode. The keypad is locked from the operator. The purpose of this mode is that if communications is lost and the shed timer times out then the controller will enter a known state of operation. The configuration of the “Shed Mode and Output” and Shed Setpoint Recall are used to configure the controller’s shed state. While in Slave Mode pushing the MAN/AUTO key enters Emergency Manual mode. The local operator then has control of the output. The controller is in Monitor Mode if the Shed timer is disabled.

10.2 Reading Control Data

Overview

The following control data can be read from the UDC3200 controller:

- Input 1
- Input 2
- PV, SP, Output

Register Addresses

Use the identifying codes listed in Table 10-1 to read the specific items.

A Write request for these codes will result in an Error message.

Table 10-1 Control Data Parameters

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Input #1	007B	123	FP	RD	In Engineering Units or Percentage
Input #2	007C	124	FP	RD	In Engineering Units or Percentage
PV, SP, Output	007A	122	FP	RD	In Engineering Units or Percentage

10.3 Read Software Options Status

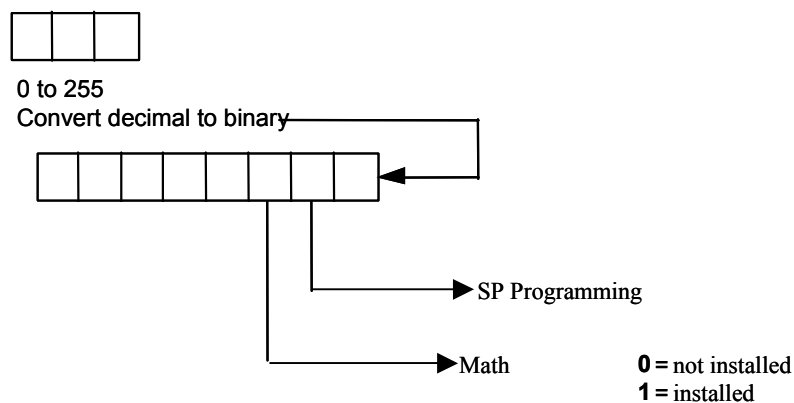
Read

Doing a Read of register address 00B9 listed in Table 10-2 will tell you which of the available options are enabled / installed or disabled / not installed.

Table 10-2 Option Status

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Option Status (Read only)	00B9	185	INT	RD	See Figure 10-1.

The data field in the response message will be a decimal number from 0 to 255. Convert the decimal number to binary as shown in Figure 10-1.to determine which options are or are not active.



EXAMPLE: 2

Binary

0	0	0	0	0	0	1	0
---	---	---	---	---	---	---	---

 SP Programming – installed

Figure 10-1 Software Option Status Information

10.4 Miscellaneous Read Onlys

10.4.1 Register Addresses for Read Onlys

The identifying register addresses listed in Table 10-3 represent some information that is Read only. No Writes allowed.

Table 10-3 Miscellaneous Read Onlys

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Software Type	009D	157	INT	RD	READ only (UDC3200) 50 = UDC3200
Software Version	00A7	167	INT	RD	READ only Value less than 255

10.4.2 SetPoint Program Read Only Information

The identifying register addresses listed in Table 10-4 represent some information for SetPoint Programming that is Read only. No Writes allowed.

Table 10-4 SetPoint Program Read Only Information

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Present SPP Segment Number	00FB	251	INT	RD	1 – 12
Segment Time Remaining in Minutes	00FC	252	INT	RD	0 – 59 Minutes
Segment Time Remaining in Hours	00FD	253	INT	RD	0 – 99 Hours
Cycles Remaining	00FE	254	INT	RD	0 – 100
Current Cycle Number	00FF	255	INT	RD	0 – 100

10.5 Setpoints

Overview

You can use two separate local setpoints in the controller. The identifying register addresses listed in Table 10-5 allow you to select which setpoint you want to use and to enter a value in Engineering Units (whichever is selected at register address 00A1) for that setpoint via communications.

Register Addresses

Make your selection using register address 00AD and enter the value for the setpoint chosen using register address in Table 10-5.

Table 10-5 Setpoint Code Selections

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Local Setpoint #1	0027	039	FP	R/W	Value within the setpoint range limits
Local Setpoint #2	0035	053	FP	R/W	Value within the setpoint range limits
Local Setpoint #3	0074	116	FP	R/W	Value within the setpoint range limits
Number of Local Setpoints	00AD	173	INT	R/W	00 = Local Setpoint #1 only 01 = 2nd Local Setpoint via keyboard or communications 03 = 3rd Local Setpoint via keyboard or communications

Associated Parameters

Refer to Table 10-6 to display or change any of the parameters associated with the setpoint.

Table 10-6 Setpoint Associated Parameters

Parameter	Register Address	
	Hex	Decimal
Setpoint Limits	0007, 0008	007, 008
Computer Setpoint	007D	125

10.6 Using a Computer Setpoint (Overriding Controller Setpoint)

Overview

You can use a setpoint generated from the computer to override the setpoint being used by the controller. The value generated by the computer will have ratio and bias applied by the controller.

Register Addresses

Use the identifying code in Table 10-7 to enter the computer setpoint.

Table 10-7 Computer Setpoint Selection

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Computer Setpoint	007D	125	FP	R/W	Value from computer with Ratio and Bias applied by the controller. Within the Setpoint Range Limits in Engineering Units or Percent.

Shed

The computer setpoint override will continue until SHED from communications occurs or the controller is placed into monitor mode through communications. Doing periodic SLAVE READS within the shed time will allow the override to continue until communication is stopped and shed time elapses. Does not apply to IR communications.

ATTENTION

0 Shed (code 79) allows the override to continue indefinitely or until the reset shed timer register address 1B90 is written using function code 6 or register address 7F using function code 21. Any data value can be written because it is ignored.

When SP is overridden, the upper display becomes “C” momentarily, and the lower display shows the CSP value as CSP XXXX.

Table 10-7.1 Shed Timer Reset

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Shed Timer Reset	007F	127	FP	W	Exit Slave Mode IR Only

Associated Parameters

Refer to Table 10-8 for the codes to display or change any of the parameters associated with the computer setpoint.

Table 10-8 Computer Setpoint Associated Parameters

Parameter	Register Address	
	Hex	Decimal
Setpoint Limits	0007, 0008	007, 008
Local Setpoint #1	0027	039
Local Setpoint #2	0035	053
Local Setpoint #3	0074	116
Local Setpoint Selection	00AD	173
Computer Setpoint Ratio	005A	90
Computer Setpoint Bias	005B	91
Shed Timer Reset	007F	127

10.7 Configuration Parameters

Overview

Listed on the next pages are the identifying codes for the parameters in the various Set-up Groups in the UDC3200 Process Controller. Most of the parameters are configurable through the hosts. Some are Read Only and are indicated as such and cannot be changed.

Reading or Writing

Do a Read or Write, depending on your requirements, using the identifying code and format code listed in the tables. The range or selection available for each range is listed in the tables.

10.7.1 Tuning

Table 10-9 lists all the register addresses and ranges or selections for the function parameters in the Set-up Group Tuning.

Table 10-9 Set-up Group – Tuning

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Gain #1 or PB Note 1	0001	001	FP	R/W	0.01 to 1000 Gain 0.1 to 1000 PB
Rate #1 Note 1	0002	002	FP	R/W	0.00 to 10.00
Reset #1 Note 1	0003	003	FP	R/W	0.02 to 50.00
Manual Reset	000D	013	FP	R/W	–100 to +100
Gain #2 or PB #2 Note 1	0004	004	FP	R/W	0.01 to 1000 Gain 0.1 to 1000 PB
Rate #2 Note 1	0005	005	FP	R/W	0.00 to 10.00
Reset #2 Note 1	0006	006	FP	R/W	0.02 to 50.00
Cycle Time #1	0015	21	INT	R/W	1 to 120 seconds
Cycle Time #2	0016	22	INT	R/W	1 to 120 seconds

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Lockout (keyboard only) Changes to data are always possible via communications regardless of this configuration.	0084	132	INT	R/W	0 = No Lockout 1 = Calibration Locked out 2 = +Configuration – Timer, Tuning, SP Ramp, Accutune are read/write 3 = +View – Tuning and SP Ramp are read/write, no other parameters are available 4 = Maximum Lockout
Security Code	0050	080	INT	R/W	0 to 9999
Man/Auto Key Lockout	00BF	191	INT	R/W	0 = Disable 1 = Enable
Run/Hold Key Lockout	00EE	238	INT	R/W	0 = Disable 1 = Enable
Setpoint Key Lockout	00ED	237	INT	R/W	0 = Disable 1 = Enable

NOTE 1: Writes to these locations are not available when Accutune is enabled.

10.7.2 SP Ramp/Rate/Program

Table 10-10 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Setpoint Ramp/Rate.

Table 10-10 Set-up Group – Setpoint Ramp/Rate

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
SP Ramp	0096	150	INT	R/W	0 = Disabled 1 = Enabled
Single SP Ramp Time	0019	25	FP	R/W	0 to 255 (minutes)
Final Ramp SP Value	001A	026	FP	R/W	PV Range in Engineering Units
SP Rate	00F0	240	INT	R/W	0 = Disabled 1 = Enabled
Rate Up (EU/HR)	006C	108	FP	R/W	0 to 9999
Rate Down (EU/HR)	006D	109	FP	R/W	0 to 9999
Setpoint Program	00B2	178	INT	R/W	0 = Disabled 1 = Enabled
Start Segment #	0058	88	FP	R/W	1 to 12
End Segment # (Soak)	00B0	176	INT	R/W	0 = Soak 2 1 = Soak 4 2 = Soak 6 3 = Soak 8 4 = Soak 10 5 = Soak 12
Engineering Units or Ramp Segments	00B6	182	INT	R/W	0 = HRS:MIN 1 = Degrees/Minute 2 = EU/Hour
Program Recycles	0059	89	FP	R/W	0 to 100
Guaranteed Soak Deviation	0057	087	FP	R/W	0 to 99.9 (0 = no soak)
Program End State	00B5	181	INT	R/W	0 = Disable SP Program 1 = Hold at Program End

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Controller Status at Program End	00B4	180	INT	R/W	0 = Last Setpoint and Mode 1 = Manual, Failsafe Output
Reset SP Program (ToBEGIN)	00B3	179	INT	R/W	0 = Disable 1 = Via Keypad
PV Hotstart	00E2	226	INT	R/W	0 = Disabled 1 = Enabled
Segment #1 Ramp Time	0039	057	FP	R/W	99.59 (0-99 Hrs:0-59 Min) or 0 to 999 (Degrees/Minute)
Segment #2 Soak Setpoint Value	003A	058	FP	R/W	Within Setpoint Limits
Segment #2 Soak Time	003B	059	FP	R/W	99.59 (0-99 Hrs:0-59 Min)
Segment #3 Ramp Time	003C	060	FP	R/W	99.59 (0-99 Hrs:0-59 Min) or 0 to 999 (Degrees/Minute)
Segment #4 Soak Setpoint Value	003D	061	FP	R/W	Within Setpoint Limits
Segment #4 Soak Time	003E	062	FP	R/W	99.59 (0-99 Hrs:0-59 Min)
Segment #5 Ramp Time	003F	063	FP	R/W	99.59 (0-99 Hrs:0-59 Min) or 0 to 999 (Degrees/Minute)
Segment #6 Soak Setpoint Value	0040	064	FP	R/W	Within Setpoint Limits
Segment #6 Soak Time	0041	065	FP	R/W	99.59 (0-99 Hrs:0-59 Min)
Segment #7 Ramp Time	0042	066	FP	R/W	99.59 (0-99 Hrs:0-59 Min) or 0 to 999 (Degrees/Minute)
Segment #8 Soak Setpoint Value	0043	067	FP	R/W	Within Setpoint Limits

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Segment #8 Soak Time	0044	068	FP	R/W	99.59 (0-99 Hrs:0-59 Min)
Segment #9 Ramp Time	0045	069	FP	R/W	99.59 (0-99 Hrs:0-59 Min) or 0 to 999 (Degrees/Minute)
Segment #10 Soak Setpoint Value	0046	070	FP	R/W	Within Setpoint Limits
Segment #10 Soak Time	0047	071	FP	R/W	99.59 (0-99 Hrs:0-59 Min)
Segment #11 Ramp Time	0048	072	FP	R/W	99.59 (0-99 Hrs:0-59 Min) or 0 to 999 (Degrees/Minute)
Segment #12 Soak Setpoint Value	0049	073	FP	R/W	Within Setpoint Limits
Segment #12 Soak Time	004A	074	FP	R/W	99.59 (0-99 Hrs:0-59 Min)

10.7.3 Accutune

Table 10-11 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Adaptive Tune.

Table 10-11 Set-up Group – Adaptive Tune

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Fuzzy Overshoot Suppression	00C1	193	INT	R/W	0 = Disabled 1 = Enabled
Accutune Enable	0098	152	INT	R/W	0 = Accutune Disabled 1 = Tune
Accutune Duplex selection	00E1	225	INT	R/W	0 = Manual 1 = Auto 2 = Disable (blend)
Accutune Error (Read only)	0097	151	INT	R/W	0 = None 3 = Process Identification failed 4 = Accutune aborted on command 5 = Running

10.7.4 Algorithm

Table 10-12 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Algorithm.

Table 10-12 Set-up Group – Algorithm

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Control Algorithm Selection (Selection here will affect ID code 160 in Output Algorithms.)	0080	128	INT	R/W	0 = ON/OFF 1 = PID-A 2 = PID-B 3 = PD-A with Manual Reset 4 = Three Position Step 5 = Disable
Timer	00D8	216	INT	R/W	0 = Disable 1 = Enable
Period	0063	099	FP	R/W	00.00 TO 99.59
Start (Initiation)	00D9	217	INT	R/W	0 = Key (Run/Hold Key) 1 = Alarm 2
LDISP (Selection)	00DA	218	INT	R/W	0 = TI REM 1 = Elapsed Time
Timer Reset	00D6	214	INT	R/W	0 = Key (Run/Hold Key) 1 = AL1 (Alarm 1 or Key)
Timer Increment	00D7	215	INT	R/W	0 = Minutes (Counts hr/minute) 1 = Sec (Counts min/sec)

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Input Algorithm 1 † Input source selected via ID 205, 206, 207.	00CC	204	INT	R/W	0 = None 1 = Weighted Average (LSP) † 2 = Feedforward – Summer † 3 = Feedforward – Multiplier † 4 = Unused 5 = Summer (with ratio and bias) † 6 = Input High Select (with ratio and bias) † 7 = Input low Select (with ratio and bias) † 8 = General Math A (sq. rt., mult., div.) † 9 = General Math B (sq. rt., mult.) † 10 = General Math C (mult., div.) † 11 = General Math D (mult.) † 12 = Carbon A 13 = Carbon B 14 = Carbon C 15 = Carbon D 16 = Carbon FCC 17 = Dewpoint 18 = Oxygen
Constant K	002D	045	FP	R/W	0.001 to 1000
Calc High	001F	031	FP	R/W	–999.0 to +9999 in Engineering Units
Calc Low	0020	032	FP	R/W	–999.0 to +9999 in Engineering Units
PV Range Low	0036	054	FP	RD	–999.0 to +9999 in Engineering Units
PV Range High	0037	055	FP	RD	–999.0 to +9999 in Engineering Units

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Input Algorithm 1 Input A Selection (used with ID 204 math calculations)	00CD	205	INT	R/W	0 = Input 1 1 = Input 2 2 = Output
Input Algorithm 1 Input B Selection (used with ID 204 math calculations)	00CE	206	INT	R/W	0 = Input 1 1 = Input 2 2 = Output
Input Algorithm 1 Input C Selection (used with ID 204 math calculations)	00CF	207	INT	R/W	0 = None 1 = Input 1 2 = Input 2 3 = Output
Algorithm1 bias	005C	092	FP	R/W	-999.0 to 9999 in Engineering Units
Percent Carbon Value	002E	046	FP	R/W	0.02 to 0.350
Percent Hydrogen	0022	034	FP	R/W	1 to 99 (% H2)

10.7.5 Output Algorithms

Table 10-13 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Output.

Table 10-13 Set-up Group – Output

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Output Algorithm	00A0	160	INT	R/W	0 = Time Simplex 1 = Current Simplex 2 = Three Position Step or Position Proportioning 3 = Time Duplex 4 = Current Duplex 5 = Current/Time Duplex 6 = Time/Current Duplex
Relay Cycle Time Increments	00BE	190	INT	R/W	0 = 1 second increments 1 = 1/3 second increments
Motor Time for Positional Proportional	004B	075	INT	R/W	5 to 1800 seconds
Relay Output Action	00F3	243	INT	R/W	0 = 1 OFF 2 OFF 1 = 1 ON 2 OFF 2 = 1 OFF 2 ON 3 = 1 ON 2 ON
Current Range for Current Duplex	0099	153	INT	R/W	0 = Full (100%) 1 = Split (50%)
Current Output (Auxiliary Output Mode)	00F2	242	INT	R/W	0 = Disable 1 = Input 1 2 = Input 2 3 = PV 4 = Deviation 5 = Output 6 = SP 7 = LSP 8 = RSP 9 = Input Algorithm 1
Current Output Range	00EA	235	INT	R/W	0 = 4-20 mA 1 = 0-20 mA

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Current Low Scaling Factor	0064	100	FP	R/W	Within the range of the selected variable in ID 242
Current High Scaling Factor	0065	101	FP	R/W	Within the range of the selected variable in ID 242

10.7.6 Input 1

Table 10-14 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Input 1.

Table 10-14 Set-up Group – Input 1

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Input 1 Type	00A8	168	INT	R/W	1 = B TC 2 = E TC H 3 = E TC L 4 = J TC H 5 = J TC M 6 = J TC L 7 = K TC H 8 = K TC M 9 = K TC L 10 = NNM H 11 = NNM L 12 = Nicrosil H TC 13 = Nicrosil L TC 14 = R TC 15 = S TC 16 = T TC H 17 = T TC L 18 = W TC H 19 = W TC L 20 = 100 PT RTD 21 = 100 PT LO RTD 22 = 200 PT RTD 23 = 500 PT RTD 24 = Radiamatic RH 25 = Radiamatic RI 26 = 0-20 mA 27 = 4-20 mA 28 = 0-10 mV 29 = 0-50 mV 30 = 100 mV 31 = 0-5 Vdc 32 = 1-5 Vdc 33 = 0-10 Vdc 34 = Unused 35 = Carbon 36 = Oxygen 37 = Thermocouple Differential
					ATTENTION Changing the Input Type will result in the loss of Field Calibration values and will restore the Factory Calibration values.

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Input 1 Transmitter Characterization	00A9	169	INT	R/W	0 = B TC 1 = E TC H 2 = E TC L 3 = J TC H 4 = J TC M 5 = J TC L 6 = K TC H 7 = K TC M 8 = K TC L 9 = NNM H 10 = NNM L 11 = Nicrosil H TC 12 = Nicrosil L TC 13 = R TC 14 = S TC 15 = T TC H 16 = T TC L 17 = W TC H 18 = W TC L 19 = 100 PT RTD 20 = 100 PT LO RTD 21 = 200 PT RTD 22 = 500 PT RTD 23 = Radiamatic RH 24 = Radiamatic RI 25 = Linear 26 = Square Root
Input 1 High Range Value	001D	029	FP	R/W	–999. to 9999. Engineering Units (Linear types only)
Input 1 Low Range Value	001E	030	FP	R/W	–999 to 9999. Engineering Units (Linear types only)
Input 1 Ratio	006A	106	FP	R/W	–20.00 to 20.00
Input 1 Bias	006B	107	FP	R/W	–999 to 9999. Engineering Units
Input 1 Filter	002A	042	FP	R/W	0 to 120 seconds

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Burnout (Open Circuit Detection)	00A4	164	INT	R/W	0 = None and Failsafe 1 = Upscale 2 = Downscale 3 = No Failsafe
Emissivity	0017	023	FP	R/W	0.01 to 1.00

10.7.7 Input 2

Table 10-15 lists all the register addresses and ranges or selections for the function parameters in Set-up Group Input 2.

Table 10-15 Set-up Group – Input 2

Parameter Description	Register Address		Data Type	Access	Data Range or Enumerated Selection
	Hex	Decimal			
Input 2 Type	00AA	170	INT	R/W	0 = Disable 1 = B TC 2 = E TC H 3 = E TC L 4 = J TC H 5 = J TC M 6 = J TC L 7 = K TC H 8 = K TC M 9 = K TC L 10 = NNM H 11 = NNM L 12 = Nicrosil H TC 13 = Nicrosil L TC 14 = R TC 15 = S TC 16 = T TC H 17 = T TC L 18 = W TC H 19 = W TC L 20 = 100 PT RTD 21 = 100 PT LO RTD 22 = 200 PT RTD 23 = 500 PT RTD 24 = Radiamatic RH 25 = Radiamatic RI 26 = 0-20 mA 27 = 4-20 mA 28 = 0-10 mV 29 = 0-50 mV 30 = 0-100 mV 31 = 0-5 Vdc 32 = 1-5 Vdc 33 = 0-10 Vdc 34 = Slidewire 35 = Unused 36 = Unused 37 = Thermocouple Differential
					ATTENTION Changing the Input Type will result in the loss of Field Calibration values and will restore the Factory Calibration values.